

transport when they make structures small enough. So far, they have built an all-graphene planar field-effect transistor.

The key to properties of the new circuitry is the width of the ribbons that confine the electrons in a quantum effect similar to that seen in carbon nanotubes. Scientists have developed and discovered many exciting properties of carbon nanotubes over the past decade and believe that the primary value of carbon nanotubes has been in calling attention to the useful properties of graphene. Carbon nanotubes conduct electricity with virtually no resistance and therefore have attracted strong interest for use in transistors and other devices.

Nanotubes are simply graphene that has been rolled into a cylindrical shape. Using narrow ribbons of graphene can give all the properties of nanotubes because those properties are due to the graphene and the confinement of the electrons, not the nanotube structures. The band structure of zig-zag type graphene nanoribbons can be semiconducting or metallic depending on the ribbon's width (chirality).

Continuous graphene circuitry can be produced using standard microelectronic processing techniques, potentially allowing creation of a roadmap for high-volume graphene electronics manufacturing. Experts envision using the graphene electronics for specialised applications, potentially within conventional silicon-based systems. Graphene systems could also be used as the foundation for molecular electronics, helping resolve resistance issues that today affect such systems.

There is a huge advantage to making a system out of one continuous material, compared to having different materials with different interfaces. Researchers expect to continue improving their materials and fabrication processes, while producing and testing new structures towards building a new class of electronics based on graphene.

However, serious obstacles must be overcome before graphene nanotube-



Kevin Brenner holds a fabricated graphene sample (Photo courtesy: Georgia Tech)

based electronic devices could be scaled up into high-volume industrial products. After some initial problems with graphene-coated lithium batteries that charge in minutes, some scientists have found another use for graphene. They have been able to make the electrons in graphene react as if they were exposed to a very strong magnetic field—something that has big implications for how we build the smallest, most basic parts of electronic devices.

Scientists were particularly excited about the control that they could potentially demonstrate over electrons around the graphene nanobubbles. The ability to control the existence of such powerful magnetic-like forces could potentially lead to the development of smaller, more powerful transistor switches in semiconductors.

Researchers at the Georgia Institute of Technology have also developed a 'template growth' technique for fabricating nanometre-scale graphene devices. The method addresses a significant obstacle to the use of this promising material in future generations of high-performance electronic devices.

The technique involves etching patterns into the silicon carbide surfaces on which epitaxial graphene is grown. The patterns serve as templates directing the growth of graphene structures, allowing the formation of nanoribbons of specific widths without the use of e-beams or other destructive cutting techniques. Graphene nanoribbons produced with these templates have

smooth edges that avoid electron-scattering problems.

The new technique has been used to fabricate an array of 10,000 top-gated graphene transistors on a 0.24cm² chip—believed to be the largest density of graphene devices reported so far.

Potential applications

Single-molecule gas detection. Graphene makes an excellent sensor due to its 2D structure. The fact that its entire volume is exposed to its surrounding makes it very efficient to detect adsorbed molecules. Molecule detection is indirect: as a gas molecule adsorbs to the surface of graphene, the location of adsorption experiences a local change in electrical resistance.

While this effect occurs in other materials too, graphene is superior due to its high electrical conductivity (even when few carriers are present) and low noise which makes the change in resistance detectable.

Graphene transistors. Graphene exhibits a pronounced response to perpendicular external electric fields, allowing one to build field-effect transistors (FETs). However, current graphene transistors show a very poor on-off ratio. Researchers are trying to find ways for improvement.

Researchers had demonstrated four different types of logic gates—each composed of a single graphene transistor—and built an experimental graphene chip known as a frequency multiplier. The chip was capable of